Macroeconomics III - Lecture 5b

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April 28th, 2026

Both "Semi-Endogenous" and "Truly Endogenous" growth?

- In the previous lecture we have seen two different growth theories:

1. Semi-endogenous growth theory - invented by Jones (*Journal of Political Economy*, 1995), Kortum (*Econometrica*,1997), and Segerstrom (*American Economic Review*, 1998)
2. Truly endogenous growth theory without scale effects - Smulders and Van de Klundert (*European Economic Review*, 1995), Dinopoulos and Thompson (*Journal of Economic Growth*, 1998), Peretto (*Journal of Economic Growth*, 1998), Young (*Journal of Political Economy*, 1998), Howitt (*Journal of Political Economy*, 1999)

- They are both empirically strong
- They are very different in their view of the innovative process and in their long-run policy implications
- What if they were both right?

Both "Semi-Endogenous" and "Truly Endogenous" growth? I

- Nobody until this year has ever attempted to join the two models
- Very recently: new models by Cozzi (2017a, *Economics Letters*) and Cozzi (2017b, *Economics Letters*):
- Simpler article: G. Cozzi, (2017a). "Endogenous Growth, Semi-Endogenous Growth... or Both? A Simple Hybrid Model", *Economics Letters*, Elsevier, vol. 154, pp. 28-30. Link: https://mpra.ub.uni-muenchen.de/77775/1/MPRA_paper_77775.pdf
- More complicated article: G. Cozzi, (2017b). "Combining Semi-Endogenous and Fully Endogenous Growth: a Generalization", *Economics Letters*, vol. 155, pp. 89-91. Link: https://mpra.ub.uni-muenchen.de/77815/1/MPRA_paper_77815.pdf
- In this class I will follow a simple version of Cozzi (2017a), calling it just Cozzi (2017) for simplicity

Cozzi Hybrid Model combining Semi- and Truly-Endogenous Growth I

- Let us assume that the true model of technological progress is a linear combination of the semi-endogenous and the truly endogenous (without scale effect) mechanisms:

$$A_{t+1} - A_t = \alpha_{\text{sem}} \rho A_t^{\phi} L_{At}^{\lambda} + (1 - \alpha_{\text{sem}}) \rho A_t s_{Rt}^{\lambda}, \tag{1}$$

  where $0 < \alpha_{\text{sem}} < 1$ is the weight of the semi-endogenous growth mechanism, while $1 - \alpha_{\text{sem}}$ is the weight of the fully endogenous growth mechanism[1]

- Hence we can express the growth rate of technology, g_t, as:

$$g_t \equiv \frac{A_{t+1} - A_t}{A_t} = \alpha_{\text{sem}} \rho A_t^{\phi - 1} L_{At}^{\lambda} + (1 - \alpha_{\text{sem}}) \rho s_{Rt}^{\lambda}. \tag{2}$$

Cozzi Hybrid Model combining Semi- and Truly-Endogenous Growth II

- This is very general as it nests the two existing growth theories as special cases:
- growth becomes semi-endogenous if $\alpha_{\text{sem}} = 1$
- growth becomes truly endogenous if $\alpha_{\text{sem}} = 0$
- all intermediate cases are possible if $0 < \alpha_{\text{sem}} < 1$
- It is not one model, but an entire class of infinitely many possible models!
- Among them one will perhaps find the actual true model of the evolution of technology
- Hence conclusions obtained with this model could contain the truth

Cozzi Hybrid Model combining Semi- and Truly-Endogenous Growth III

Note for readers of the original article: Cozzi's (2017a) original model is written in continuous time. Hence instead of writing innovation in terms of finite differences (that is $A_{t+1} - A_t$) it writes them in terms of derivatives with respect to time (that is $\dot{A}(t)$). The interpretation is the same as here. Anyway, the current lecture note is enough for the exam preparation

Cozzi Hybrid Model combining Semi- and Truly-Endogenous Growth IV

- What are the predictions of this model for the future of the world percapita GDP?
- The reader may be led into thinking that in the long-run (with constant $s_{Rt} = s_R$) the growth rate of equation (1) would tend to a combination of the semi-endogenous growth rate

$$g^{se} = (1+n)^{\frac{\lambda}{1-\phi}} - 1 \simeq \frac{\lambda n}{1-\phi}$$

and the fully endogenous growth rate

$$g^{e} \equiv \rho s_R^{\lambda}, \tag{3}$$

that is:

$$g_t \longrightarrow \alpha_{\text{sem}} \rho \left[(1+n)^{\frac{\lambda}{1-\phi}} - 1 \right] + (1-\alpha_{\text{sem}}) \rho s_R, \tag{4}$$

but this would be **wrong**!

Cozzi Hybrid Model combining Semi- and Truly-Endogenous Growth V

- Instead, Cozzi (2017) proves that the combined model equation (1) implies that the long-run growth rate of technology will be **either** only that of the semi-endogenous growth model, $(1+n)^{\frac{\lambda}{1-\phi}} - 1$, **or** the weighted fully endogenous $(1-\alpha_{\text{sem}})\rho s_R^{\lambda}$, depending on how fast population grows.
- In general, we will call the steady state technological growth rate of our hybrid model just g^h, where superscript h stands for "hybrid".

[1] Notice that α_{sem} has nothing to do with the exponent α of the Cobb-Douglas production function!

The Complete Cozzi (2017) Model I

$$Y_t = K_t^{\alpha}(A_t L_{Yt})^{1-\alpha}$$

$$A_{t+1} - A_t = \alpha_{\text{sem}} \rho A_t^{\phi} L_{At}^{\lambda} + (1 - \alpha_{\text{sem}}) A_t s_{Rt}^{\lambda}, \tag{5}$$

$$K_{t+1} = sY_t + (1 - \delta) K_t$$

$$L_{t+1} = (1 + n) L_t$$

$$L_{Yt} + L_{At} = L_t$$

$$L_{At} = s_R L_t$$

- Parameters: α, α_{sem}, ρ, ϕ, λ, δ, s, n, and s_R.
- Notice that s and s_R can be affected by policy
- State variables: K_t, L_t, and A_t.

Implications I

- The technological growth part of the model can be solved separately
- Given the resulting technological growth rate, g_t, and its steady state level, g^h, the rest of the model follows the same steps of the previous lecture note (as in Chapter 9):
- define $\tilde{k}_t \equiv K_t / A_t L_t$ and $\tilde{y}_t \equiv Y_t / A_t L_t$, which as in Chapter 9 gives $\tilde{y}_t = \tilde{k}_t^{\alpha}(1 - s_R)^{1-\alpha}$
- The transition of $\tilde{k}_t$ follows exactly that of Chapter 9:

$$\tilde{k}_{t+1} = \frac{1}{(1+n)(1+g_t)} \left[s\tilde{k}_t^{\alpha}(1 - s_R)^{1-\alpha} + (1-\delta)\tilde{k}_t \right] \tag{6}$$

Implications II

- As in Chapter 9, $\tilde{k}_t$ will converge to its positive steady state value

$$\tilde{k}^* = \left(\frac{s}{n + g^h + \delta + ng^h} \right)^{\frac{1}{1-\alpha}} (1 - s_R)$$

and $\tilde{y}_t$ will converge to its positive steady state value

$$\tilde{y}^* = \left(\frac{s}{n + g^h + \delta + ng^h} \right)^{\frac{\alpha}{1-\alpha}} (1 - s_R)$$

- Consequently, percapita output $y_t = Y_t / L_t = \tilde{y}_t A_t$ will eventually grow at the same rate as technology A_t, which we will obtain shortly.

A very simple case: constant population I

- Let us assume $n = 0$, that is population is constant: $L_t = L$
- Then also R&D employment will be constant at $L_{At} = s_R L_t \equiv L_A$
- Our growth equation becomes:

$$g_t = \alpha_{\text{sem}} \rho A_t^{\phi-1} L_A^{\lambda} + (1 - \alpha_{\text{sem}}) \rho s_R^{\lambda}. \tag{7}$$

- Notice that g_t is growing at a positive rate no smaller than $(1 - \alpha_{\text{sem}}) \rho s_R^{\lambda} > 0$
- Hence A_t will grow without bound as time tends to infinity
- As a consequence $A_t^{\phi-1} = \frac{1}{A_t^{1-\phi}} \underset{t\to\infty}{\longrightarrow} 0$
- Hence growth rate equation (7) will tend to

$$g_t \underset{t\to\infty}{\longrightarrow} 0 + (1 - \alpha_{\text{sem}}) \rho s_R^{\lambda} = (1 - \alpha_{\text{sem}}) g^e. \tag{8}$$

- Therefore we can conclude that *with constant population the long-run growth rate of technology, and therefore that of percapita GDP, will be fully endogenous*

Another simple case: shrinking population I

- Let us assume $n < 0$, that is population is declining: $L_t \underset{t\to\infty}{\longrightarrow} 0$
- Then also R&D employment will be shrinking $L_{At} = s_R L_t \underset{t\to\infty}{\longrightarrow} 0$
- but then $A_t^{\phi-1} L_{At}^{\lambda} \underset{t\to\infty}{\longrightarrow} 0$ as well
- As a consequence, also in this case, our growth rate equation (7) will tend to

$$\begin{aligned} g_t &= \alpha_{\text{sem}} \rho A_t^{\phi-1} L_{At}^{\lambda} + (1-\alpha_{\text{sem}})\rho s_R^{\lambda} \underset{t\to\infty}{\longrightarrow} 0 + (1-\alpha_{\text{sem}})\rho s_R^{\lambda} && (9) \\ &= (1-\alpha_{\text{sem}}) g^e && (10) \end{aligned}$$

- Therefore we can conclude that *with shrinking population the long-run growth rate of technology, and therefore that of percapita GDP, will be fully endogenous.*

The case of growing population I

- Let us finally assume $n > 0$, that is that population is increasing: $L_t \underset{t\to\infty}{\longrightarrow} \infty$
- We can rewrite our growth equation as

$$g_t - (1-\alpha_{\text{sem}})\rho s_R^{\lambda} = \alpha_{\text{sem}}\rho A_t^{\phi-1} L_{At}^{\lambda}$$

- Rewriting it also for $t+1$ gives

$$g_{t+1} - (1-\alpha_{\text{sem}})\rho s_R^{\lambda} = \alpha_{\text{sem}}\rho A_{t+1}^{\phi-1} L_{At+1}^{\lambda}$$

- Dividing each side of the second equation by the corresponding side of the first equation, we obtain:

$$\begin{aligned}\frac{g_{t+1} - (1-\alpha_{\text{sem}})\rho s_R^{\lambda}}{g_t - (1-\alpha_{\text{sem}})\rho s_R^{\lambda}} &= \frac{\alpha_{\text{sem}}\rho A_{t+1}^{\phi-1} L_{At+1}^{\lambda}}{\alpha_{\text{sem}}\rho A_t^{\phi-1} L_{At}^{\lambda}} = \frac{A_{t+1}^{\phi-1} s_R^{\lambda} L_{t+1}^{\lambda}}{A_t^{\phi-1} s_R^{\lambda} L_t^{\lambda}} = \\ &= (g_t+1)^{\phi-1}(1+n)^{\lambda}\end{aligned}$$

The case of growing population II

- In a balanced growth path (steady state) $g_{t+1} = g_t = g^h$ and hence the first term is just 1
- Hence we can solve $1 = \left(g^h + 1\right)^{\phi - 1} (1+n)^{\lambda}$ for g^h and get

$$g^h = (1+n)^{\frac{\lambda}{1-\phi}} - 1 \equiv g^{se}$$

- The same growth rate as predicted by **semi-endogenous growth** theory!
- This result holds only if $(1+n)^{\frac{\lambda}{1-\phi}} - 1 \equiv g^{se} > (1-\alpha_{\text{sem}})\rho s_R^{\lambda}$
- If instead $(1-\alpha_{\text{sem}})\rho s_R^{\lambda} > g^{se} \equiv (1+n)^{\frac{\lambda}{1-\phi}} - 1$, then A_t will grow faster than $L_{At}^{\frac{\lambda}{1-\phi}}$, which implies $A_t^{\phi-1} L_{At}^{\lambda} \underset{t \to \infty}{\longrightarrow} 0$, and hence the steady state growth rate will be equal to $(1-\alpha_{\text{sem}})\rho s_R^{\lambda}$.

The case of growing population III

- Therefore the semi-endogenous growth rate g^{se} will prevail if and only if the population growth rate, n, is higher than a threshold level, $\bar{n}$, obtained by solving

$$(1+\bar{n})^{\frac{\lambda}{1-\phi}} - 1 = (1-\alpha_{\text{sem}})\rho s_R^{\lambda},$$

that is if and only if:

$$n > \bar{n} \equiv \left[(1-\alpha_{\text{sem}})\rho s_R^{\lambda} + 1\right]^{\frac{1-\phi}{\lambda}} - 1 \qquad (11)$$

- Hence *if population grows at a rate higher than* $\bar{n}$ *the growth rate of percapita GDP will converge to the semi-endogenous growth rate* g^{se}, *while if population grows at a rate lower than* $\bar{n}$ *the growth rate of percapita GDP will converge to the fully endogenous growth rate* g^{e}.

Summing Up Cozzi (2017) I

- In the long-run, per-capita GDP will follow the predictions of the **Semi-endogenous growth** theory if population growth rate n is larger than a positive threshold $\bar{n}$
- In the long-run, per-capita GDP will follow the predictions of the **fully endogenous growth** theory without scale effects if population growth rate n is lower than $\bar{n}$.
- This is irrespective of the value of α_{sem}: the world technology could be 99% semi-endogenous, and yet if the world population does not eventually tend to infinity the fully endogenous growth predictions will eventually prevail
- Viceversa, the world technology could be 99% fully endogenous, and yet if the total population in the planet keeps growing relatively fast, the semi-endogenous growth predictions will eventually prevail

Summing Up Cozzi (2017) II

- Therefore demography dictates which growth mechanism will eventually prevail
- Equation (11) shows that demography does not discrimitate between the two theories in isolation from other variables
- Most importantly, $\bar{n}$ depends on s_R
- Policy, by affecting s_R, could make the growth mechanism shift from semi-endogenous to fully endogenous!
- For example, for often used parameter values, such as $\rho = 1$, $\phi = 0.5$, $\lambda = 0.9$, $n = 0.01$, and a neutral guess $\alpha_{\text{sem}} = 0.5$, increasing the R&D employment ratio s_R from 1% to 3% - as recommended by the European Union Lisbon Strategy - would increase the threshold $\bar{n}$ from 0.44% - which makes $n > \bar{n}$ and hence steady state growth semi-endogenous - to 1.18% - which makes $n < \bar{n}$ and hence steady state growth fully endogenous
- Policy could make policy itself more effective for growth!

Summing Up Cozzi (2017) III

- If the world technology followed the hybrid class of models of Cozzi's (2017), it is possible that in the ancient times of a Malthusian stage of the world, population growth was so slow that fully endogenous growth would be dominant for long
- from the first industrial revolution on, when world population had started to grow faster, the semi-endogenous growth would have prevailed
- in more recent times, as more and more countries are experiencing a demographic transition to lower population growth, the fully endogenous growth would have more chances to re-emerge
- Very long term prediction of this model: if in the future the world is not going to be populated by an unboundedly large amount of humans, fully endogenous growth (without scale effect) will dictate the long run.